FEAT-ZD-48600 · DE-03 Database changes · Completed database changes

Data Engineering agent · item task.db_changes · source hobasa-data-engineering-agent/outputs/FEAT-ZD-48600/task.db_changes.md · approved by Data engineer sim.data_engineer

task.db_changes: FEAT-ZD-48600

Completed database changes for FEAT-ZD-48600 (ZD-48600, "Show which overdue invoices belong to vendors that usually pay late"). This is DE-03 Database changes (attempt 3).

Result: no database change is required for this feature, so none is outstanding. The step is complete with nothing to apply.

Field	Value	Source
Trace ID	FEAT-ZD-48600	record:brief
Step	DE-03 Database changes (attempt 3)	record:brief
Workbook action	Check the required database changes are completed	record:brief
Owner / gate	Data engineer / automated checks, independent judge, sampled human evaluation	record:brief
Inputs	feat.tasks (b0fd4d2), feat.design (722a98f)	record:brief
Data Engineering tasks in scope	TSK-01 (STORY-01, `paid_date` in `to_model`), TSK-02 (STORY-02, `delays_by_vendor`)	feat.tasks (section 1)
Repository and runtime	ap_pipeline, Python 3.9 standard library only	feat.tasks (section 1); feat.design (DD-1, B1)

1. What the design says about storage

Check	Finding	Source
Storage in scope	The design's data boundary covers one new record key `paid_date`, one new reject reason `'paid_date'`, an in-memory delay history, risk map and overdue rows, and CT-1 JSON. It lists "Storage, database, schema migration" as **out of this design**	feat.design (A1, Boundary "Data")
New technology or storage	DD-1: no new repository, service, runtime, library or storage	feat.design (A7, DD-1)
Purity of new functions	NF-10: every new function is pure and in-memory, with no I/O, no network and no mutation of inputs	feat.design (A5, NF-10)
Platform trigger	Adding storage or a network call from the libraries would reopen DES-01; the design triggers none of these	feat.design (A8)
Invoice record shape	DC-1 is a plain Python `dict`, not a class; `paid_date` is added as a key (a `date` or `None`)	feat.design (A3, DC-1; B1.3)
Delay history	DC-2 is an in-memory `Dict[str, List[int]]`	feat.design (A3, DC-2; B1.3)

2. What the Data Engineering tasks change

Task	Files in budget	Database object touched	Source
TSK-01	`src/ap_pipeline/transform.py`; `src/ap_pipeline/__init__.py` (version step only); `tests/test_pipeline.py`	None	feat.tasks (TSK-01)
TSK-02	New `src/ap_pipeline/delays.py`; new `tests/test_delays.py`	None	feat.tasks (TSK-02)

The "must not change" list for TSK-01 keeps `ingest.py`, `dq.py`, `pipeline.py` and `schedule.json` unchanged (feat.tasks TSK-01; feat.design B1.1). No task in feat.tasks lists a DDL file, a migration script or a table.

3. What the repository holds

Item	Finding	Database access	Source
Repository	ap_pipeline has no database; no migrations exist or are required	None	tool:repo-access
`ingest.parse_feed`, `dq.quality_report`, `pipeline.run`	Unchanged; the new reject reason `'paid_date'` reaches `reject_reasons` with no change to them	None added	feat.design (HC-3, B1.1)
Feed row (input)	A `dict` of `str` from `csv.DictReader`. The `paid_date` key may be missing (feed v2) or `None` (a short row)	None: the feed is CSV text	feat.design (B1.3)
`transform.to_model`	It builds each record in an existing `try` block and rejects rows on `(KeyError, ValueError, InvalidOperation, AttributeError)`; TSK-01 adds the `paid_date` parse after that block	None	feat.design (B1.2); feat.tasks (TSK-01)
`schedule.json`	No change (daily, 02:00, Asia/Kolkata)	None	feat.design (B1.1)
`__init__.py`	Package version `0.4.0` today; version step only	None	feat.design (B1); feat.tasks (section 2)

Feed v2, which has no `paid_date` column, must still load, and TSK-01 handles the optional column inside `to_model` (feat.tasks TSK-01; feat.design NF-4, B1.4).

4. Database change register

ID	Change	Required	Status	Source
—	No database, table, column, index, view or migration change	No	Not applicable: nothing to apply or roll back	feat.design (A1, DD-1, NF-10); feat.tasks (TSK-01, TSK-02); tool:repo-access

A rollback plan, migration scripts, an environment promotion order and post-change validation queries do not apply.

5. Effect on later Data Engineering steps

- DE-05 (Data Model Changes): the logical Invoice record gains `paid_date` (a `date` or `None`), and there is a new reject reason `'paid_date'`. These change the registered data model, not a physical database, so they belong to DE-05, not to this step (feat.design B1.3, DC-1).
- DE-08 (Coding): this step adds no database prerequisite to TSK-01 or TSK-02.

6. Decisions and memory checked

Decisions D-2026-031, D-2026-044 and D-2026-050 and lessons L-007 and L-011 cover day counts, Decimal amounts and the exclusion of paid invoices. None of them needs storage, so none conflicts with a no-database outcome (record:memory).

7. Open questions (non-blocking)

ID	Question	Interim answer	Owner (proposed)
DBQ-1	Carried from feat.design DQ-1: no input names the production component that calls the stages and carries CT-1. A database change would be needed only if that host later stored records, overdue rows or risk levels	None in this feature. Adding storage would reopen DES-01 (feat.design A8)	Architect

8. Validation check

Check	Result	Source
Required database changes identified	None required	feat.design (A1, DD-1); tool:repo-access
Required database changes completed	Complete (nothing to apply)	tool:repo-access
No production or source-system data used	Pass: only the step's inputs and the tool view were used; no source system was queried	record:brief
Conflict with an approved output	None	feat.design; feat.tasks
Blocking questions	None	—

[Tool result] [ap_pipeline] No database in this repository's architecture; no migrations required